Mathematical Foundations of Multi-Strategy Cryptocurrency Portfolio Optimization

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Abstract

This paper presents the complete mathematical framework underlying a modular cryptocurrency portfolio optimization system. We formalize seven allocation strategies—Equal Weight, Markowitz Mean-Variance Optimization, GARCH-Enhanced Global Minimum Variance, Hierarchical Risk Parity, Equal Risk Contribution, Mean-CVaR, Black-Litterman with on-chain signal views, and Regime-Aware allocation—alongside the volatility modeling (GJR-GARCH), regime detection (Gaussian Hidden Markov Models), risk analytics, and walk-forward backtesting methodology that support them. Each method is presented with full derivations, parameter definitions, and algorithmic descriptions. This document serves as the mathematical source of truth for all project deliverables.

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1 Introduction

Cryptocurrency markets present unique challenges for portfolio construction: 24/7 continuous trading, extreme volatility, fat-tailed return distributions, and rapidly shifting correlation structures. Classical portfolio theory, developed for equity markets with moderate volatility and approximately Gaussian returns, requires careful adaptation.

This paper formalizes the mathematical methods implemented in our multi-strategy portfolio optimization dashboard. The system integrates seven distinct allocation strategies, each grounded in a different branch of quantitative finance, and supports them with advanced volatility forecasting, regime detection, on-chain signal integration, and rigorous walk-forward backtesting.

Notation. Throughout this paper, we use the following conventions:

- N denotes the number of assets in the investment universe.
- T denotes the number of time periods (daily observations).
- $\boldsymbol{w} = (w_1, \ldots, w_N)^\top \in \mathbb{R}^N$ is the portfolio weight vector.
- $\boldsymbol{\mu} \in \mathbb{R}^N$ is the vector of expected asset returns.
- $\boldsymbol{\Sigma} \in \mathbb{R}^{N \times N}$ is the covariance matrix of asset returns.
- $\mathbf{1} = (1, \ldots, 1)^\top \in \mathbb{R}^N$ is the vector of ones.
- $r_{i,t}$ denotes the log return of asset i at time t.
- All returns are daily; annualization uses a factor of 365 (crypto convention).

2 Return Modeling and Covariance Estimation

2.1 Logarithmic Returns

Definition 2.1 (Log Return)**.** *The continuously compounded (logarithmic) return of asset i at time t is*

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) = \ln P_{i,t} - \ln P_{i,t-1}, \tag{1}$$

where $P_{i,t}$ is the closing price of asset i at time t.

Log returns possess two key properties that justify their use in portfolio analytics:

1. **Time additivity.** The multi-period log return is the sum of single-period log returns:
$$r_{i,\,t:t+k} = \sum_{s=1}^{k} r_{i,\,t+s}. \tag{2}$$
2. **Approximate normality.** By the Central Limit Theorem, if daily log returns are i.i.d. with mean μ and variance σ^2, the k-period return converges to $\mathcal{N}(k\mu,\ k\sigma^2)$. In practice, cryptocurrency returns exhibit significant excess kurtosis and negative skewness (see Section 12), making the Gaussian approximation unreliable for tail risk estimation. This motivates the use of Student-t innovations in the GARCH model (Section 5) and CVaR-based optimization (Section 8).

2.2 Annualization

Since cryptocurrency markets operate continuously (365 days per year), we annualize as follows:

$$\mu_{\text{ann}} = \bar{r} \times 365, \qquad \sigma_{\text{ann}} = \hat{\sigma} \times \sqrt{365}, \tag{3}$$

where $\bar{r}$ is the sample mean daily return and $\hat{\sigma}$ is the sample daily standard deviation.

2.3 Sample Covariance Matrix

Definition 2.2 (Sample Covariance)**.** *Given the $T \times N$ matrix of centered returns, the sample covariance estimator is*

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{T-1} \sum_{t=1}^{T} (\boldsymbol{r}_t - \bar{\boldsymbol{r}})(\boldsymbol{r}_t - \bar{\boldsymbol{r}})^\top, \tag{4}$$

where $\boldsymbol{r}_t \in \mathbb{R}^N$ is the cross-sectional return vector at time t and $\bar{\boldsymbol{r}} = \frac{1}{T}\sum_{t=1}^{T} \boldsymbol{r}_t$.

When $T < 3N$—common in crypto with 50 assets and 2 years of daily data—the sample covariance is poorly conditioned and requires regularization.

2.4 Ledoit–Wolf Shrinkage Estimator

Ledoit and Wolf [17] proposed shrinking the sample covariance toward a structured target:

$$\hat{\boldsymbol{\Sigma}}_{\mathrm{LW}} = (1 - \alpha^*)\, \hat{\boldsymbol{\Sigma}} + \alpha^*\, \boldsymbol{F}, \tag{5}$$

where $\boldsymbol{F}$ is the shrinkage target and $\alpha^* \in [0, 1]$ is the optimal shrinkage intensity derived analytically.

For the *constant-correlation target* [18]:

$$F_{ij} = \begin{cases} \hat{\sigma}_{ii} & \text{if } i = j, \\ \bar{\rho}\sqrt{\hat{\sigma}_{ii}\,\hat{\sigma}_{jj}} & \text{if } i \neq j, \end{cases} \tag{6}$$

where $\hat{\sigma}_{ii} = [\hat{\boldsymbol{\Sigma}}]_{ii}$ denotes the i-th diagonal element (asset variance) of the sample covariance and $\bar{\rho}$ is the average sample correlation. The optimal intensity minimizes the expected Frobenius loss $\mathbb{E}\Big[\|\hat{\boldsymbol{\Sigma}}_{\mathrm{LW}} - \boldsymbol{\Sigma}\|_F^2\Big]$ and is given in closed form in [17].

3 Markowitz Mean-Variance Optimization

3.1 Problem Formulation

The foundational portfolio selection model of Markowitz [1] minimizes portfolio variance subject to a target expected return.

Definition 3.1 (Mean-Variance Problem)**.** *For a target return μ_0:*

$$\boxed{\begin{aligned} \min_{\boldsymbol{w}} \quad & \boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w} \\ s.t. \quad & \boldsymbol{w}^\top \boldsymbol{\mu} = \mu_0, \\ & \boldsymbol{w}^\top \mathbf{1} = 1. \end{aligned}} \tag{7}$$

3.2 Analytical Solution and the Efficient Frontier

The Lagrangian of (7) is

$$\mathcal{L}(\boldsymbol{w}, \lambda_1, \lambda_2) = \boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w} + \lambda_1\Big(\boldsymbol{w}^\top \boldsymbol{\mu} - \mu_0\Big) + \lambda_2\Big(\boldsymbol{w}^\top \mathbf{1} - 1\Big). \tag{8}$$

The following analytical results assume $\boldsymbol{\Sigma} \succ 0$ (positive definite). When $\boldsymbol{\Sigma}$ is only positive semi-definite, the inverse does not exist and numerical solvers with regularization must be used (see Section 2.4).

Setting $\nabla_{\boldsymbol{w}} \mathcal{L} = \mathbf{0}$ yields:

$$\boldsymbol{w}^* = -\frac{1}{2}\, \boldsymbol{\Sigma}^{-1}\Big(\lambda_1\, \boldsymbol{\mu} + \lambda_2\, \mathbf{1}\Big). \tag{9}$$

Define the scalar constants:

$$A = \mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}, \quad B = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}, \quad C = \mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \mathbf{1}, \quad D = BC - A^2. \tag{10}$$

The optimal weights are a linear function of the target return:

$$\boldsymbol{w}^*(\mu_0) = \boldsymbol{g} + \boldsymbol{h}\, \mu_0, \tag{11}$$

where

$$\boldsymbol{g} = \frac{1}{D}\Big(B\, \boldsymbol{\Sigma}^{-1}\mathbf{1} - A\, \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}\Big), \quad \boldsymbol{h} = \frac{1}{D}\Big(C\, \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - A\, \boldsymbol{\Sigma}^{-1}\mathbf{1}\Big). \tag{12}$$

The **efficient frontier** in (σ_p, μ_p) space is the upper branch of the hyperbola:

$$\sigma_p^2 = \frac{1}{D}\Big(C\, \mu_p^2 - 2A\, \mu_p + B\Big). \tag{13}$$

Remark 3.1 (Two-Fund Separation)**.** *Equation (11) implies that every efficient portfolio is a linear combination of any two distinct efficient portfolios—the Two-Fund Separation Theorem.*

3.3 Global Minimum Variance Portfolio

The GMV portfolio minimizes variance without a return constraint:

$$\boldsymbol{w}_{\text{GMV}} = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}}, \tag{14}$$

with $\mu_{\text{GMV}} = A/C$ and $\sigma^2_{\text{GMV}} = 1/C$.

3.4 Maximum Sharpe Ratio (Tangency) Portfolio

Given a risk-free rate r_f, the tangency portfolio maximizes the Sharpe ratio:

$$\boldsymbol{w}_{\tan} = \frac{\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f\, \mathbf{1})}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f\, \mathbf{1})}. \tag{15}$$

The maximum attainable Sharpe ratio is:

$$S_{\max} = \sqrt{(\boldsymbol{\mu} - r_f\, \mathbf{1})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f\, \mathbf{1})}. \tag{16}$$

3.5 Weight Constraints

In practice, we impose box constraints to prevent concentration:

$$0 \leq w_i \leq w_{\max} \quad \forall\, i = 1, \ldots, N, \qquad \sum_{i=1}^{N} w_i = 1. \tag{17}$$

With these constraints, the closed-form solutions (14)–(15) no longer hold, and the problem is solved numerically via quadratic programming (CLARABEL solver).

4 Equal Weight Baseline

The naïve diversification strategy assigns equal capital to all assets:

$$w_i = \frac{1}{N}, \quad i = 1, \ldots, N. \tag{18}$$

DeMiguel, Garlappi, and Uppal [12] showed that no optimized strategy consistently outperforms $1/N$ out-of-sample across 7 empirical datasets. The sample size required for MVO to dominate $1/N$ scales as:

$$T^* \propto \frac{N \cdot \sigma^2}{\mu^2}, \tag{19}$$

which, calibrated to financial data, yields $T^* \approx 3{,}000$ months (approximately 250 years) for $N = 25$ assets. With typical crypto data horizons of 2–5 years, this threshold is far from met, making $1/N$ a strong baseline.

5 GJR-GARCH Volatility Forecasting

5.1 Model Specification

The GJR-GARCH(1,1,1) model [5] captures the *leverage effect*—the empirical observation that negative shocks increase future volatility more than positive shocks of equal magnitude.

Definition 5.1 (GJR-GARCH(1,1,1))**.**

$$r_t = \mu + \varepsilon_t, \tag{20}$$

$$\varepsilon_t = \sigma_t\, z_t, \qquad z_t \sim t_\nu(0, 1), \tag{21}$$

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \gamma\, \varepsilon_{t-1}^2\, \mathbb{I}(\varepsilon_{t-1} < 0) + \beta\, \sigma_{t-1}^2, \tag{22}$$

where $\mathbb{I}(\cdot)$ is the indicator function, and t_ν denotes the standardized Student-t distribution with $\nu > 2$ degrees of freedom.

Parameters.

- $\omega > 0$: baseline variance floor.
- $\alpha \geq 0$: ARCH coefficient (symmetric shock impact).
- $\gamma \geq 0$: leverage coefficient (additional impact for $\varepsilon_{t-1} < 0$).
- $\beta \geq 0$: GARCH coefficient (variance persistence).
- $\nu > 2$: degrees of freedom of the Student-t innovation distribution.

Remark 5.1. *The impact of a negative shock is $\alpha + \gamma$, while the impact of a positive shock is α. The ratio $(\alpha+\gamma)/\alpha$ quantifies the* asymmetry ratio *when $\alpha > 0$. When $\alpha = 0$ (no symmetric ARCH effect), asymmetry is driven entirely by γ, and the model reduces to a pure leverage specification.*

5.2 Stationarity and Unconditional Variance

Proposition 5.1 (Stationarity Condition)**.** *The GJR-GARCH(1,1,1) process is covariance-stationary if and only if*

$$\kappa := \alpha + \frac{\gamma}{2} + \beta < 1. \tag{23}$$

The factor $\gamma/2$ arises because $\mathbb{E}[\mathbb{I}(\varepsilon_{t-1} < 0)] = 1/2$ under a symmetric distribution. The unconditional variance is:

$$\bar{\sigma}^2 = \frac{\omega}{1-\kappa} = \frac{\omega}{1-\alpha-\gamma/2-\beta}. \tag{24}$$

5.3 Maximum Likelihood Estimation

Under the Student-t distributional assumption, the per-observation log-likelihood is:

$$\ell_t = \ln\Gamma\Big(\frac{\nu+1}{2}\Big) - \ln\Gamma\Big(\frac{\nu}{2}\Big) - \frac{1}{2}\ln\Big(\pi(\nu-2)\Big) - \frac{1}{2}\ln\sigma_t^2 - \frac{\nu+1}{2}\ln\left(1 + \frac{\varepsilon_t^2}{\sigma_t^2(\nu-2)}\right), \tag{25}$$

where $\Gamma(\cdot)$ is the gamma function. The parameters $\boldsymbol{\theta} = (\mu, \omega, \alpha, \gamma, \beta, \nu)$ are estimated by maximizing $\ell(\boldsymbol{\theta}) = \sum_{t=1}^{T} \ell_t$ subject to the non-negativity and stationarity constraints.

5.4 Input Rescaling

For numerical stability, daily log returns (typically of order 10^{-2}) are multiplied by 100 before fitting:

$$y_t = 100 \cdot r_t. \tag{26}$$

After fitting, the conditional volatility is converted back: $\hat{\sigma}_t = \hat{\sigma}_t^{(\text{scaled})}/100$, and variance forecasts are divided by 10,000.

5.5 Multi-Step Ahead Forecasting

The one-step forecast is computed directly from (22). For $h \geq 2$ steps ahead, using the law of iterated expectations:

$$\sigma_{t+h|t}^2 = \bar{\sigma}^2 + \kappa^{h-1}\Big(\sigma_{t+1|t}^2 - \bar{\sigma}^2\Big). \tag{27}$$

This shows that forecasts mean-revert exponentially to the unconditional variance $\bar{\sigma}^2$ at rate κ.

5.6 GARCH-Modified Covariance Matrix

Following the CCC-GARCH framework of Bollerslev [6], we construct a forward-looking covariance matrix:

1. Fit GJR-GARCH(1,1,1) to each asset i independently $\rightarrow$ obtain $\hat{\sigma}_i$ (one-step-ahead forecast).
2. Compute standardized residuals: $z_{i,t} = \varepsilon_{i,t}/\sigma_{i,t}$.
3. Estimate the constant correlation matrix: $\boldsymbol{R} = \mathrm{Corr}(\boldsymbol{z}_t)$.
4. Construct the diagonal volatility matrix: $\boldsymbol{D} = \mathrm{diag}(\hat{\sigma}_1, \ldots, \hat{\sigma}_N)$.
5. Form the covariance matrix:

$$\hat{\boldsymbol{H}} = \boldsymbol{D}\,\boldsymbol{R}\,\boldsymbol{D}, \qquad \text{i.e.,} \quad \hat{H}_{ij} = \hat{\sigma}_i\,\rho_{ij}\,\hat{\sigma}_j. \tag{28}$$

This matrix replaces the sample covariance in the GARCH-Enhanced GMV strategy, providing a volatility-regime-aware estimate.

6 Hierarchical Risk Parity

Hierarchical Risk Parity (HRP), introduced by López de Prado [7], is a graph-theoretic alternative to MVO that avoids matrix inversion entirely, providing robustness to estimation error.

6.1 Step 1: Tree Clustering

Define a distance metric that converts correlation into Euclidean distance:

$$d(i,j) = \sqrt{\frac{1-\rho_{ij}}{2}}, \tag{29}$$

where ρ_{ij} is the Pearson correlation between assets i and j. Note that $d \in [0,1]$, with $d = 0$ for perfectly correlated assets and $d = 1$ for perfectly anti-correlated assets. For uncorrelated assets ($\rho = 0$), $d = 1/\sqrt{2} \approx 0.707$. Agglomerative hierarchical clustering (Ward linkage) is applied to the distance matrix to produce a dendrogram.

6.2 Step 2: Quasi-Diagonalization

The rows and columns of $\boldsymbol{\Sigma}$ are reordered according to the dendrogram's leaf ordering, producing a permuted covariance matrix $\boldsymbol{\Sigma}_{\mathrm{QD}}$ with a quasi-block-diagonal structure. This ensures that subsequent bisection splits separate dissimilar clusters.

6.3 Step 3: Recursive Bisection

Algorithm 1 HRP Recursive Bisection

Require: Covariance matrix $\boldsymbol{\Sigma}$, ordered asset indices $\mathcal{L} = [\ell_1, \ldots, \ell_N]$
Ensure: Weight vector $\boldsymbol{w}$ with $\sum_i w_i = 1$

```
Initialize w_i ← 1 for all i
Q ← {L}                                          ▷ Queue of clusters to process
while ∃ C ∈ Q with |C| > 1 do
    Split C into left half C_1 and right half C_2
    Compute inverse-variance weights within each cluster:
        w̃_k^(C_j) = (1/Σ_kk) / Σ_{m∈C_j} (1/Σ_mm)  for k ∈ C_j
    Compute cluster variance:
        V(C_j) = w̃^(C_j)ᵀ Σ_{C_j} w̃^(C_j)  for j ∈ {1, 2}
    Compute splitting factor:
        α = 1 − V(C_1) / (V(C_1) + V(C_2))                    (30)
    Update weights: w_i ← w_i · α for i ∈ C_1;   w_i ← w_i · (1 − α) for i ∈ C_2
end while
return w
```

Remark 6.1 (Robustness)**.** *HRP never inverts $\boldsymbol{\Sigma}$—it only reads diagonal entries and computes quadratic forms. The condition number $\kappa(\boldsymbol{\Sigma}) = |\lambda_{\max}|/|\lambda_{\min}|$, which amplifies estimation error in MVO by a factor of κ upon inversion, is irrelevant for HRP. This makes HRP particularly suitable for the high-dimensional, high-correlation crypto setting.*

Our implementation uses riskfolio-lib's `HCPortfolio` class, which extends the original algorithm with configurable codependence measures (Pearson, Spearman, Kendall), covariance estimators, and risk measures for the bisection step.

7 Equal Risk Contribution (Risk Parity)

7.1 Risk Decomposition

Since $\sigma(\boldsymbol{w}) = \sqrt{\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w}}$ is positively homogeneous of degree 1 in $\boldsymbol{w}$, Euler's theorem gives the decomposition:

$$\sigma(\boldsymbol{w}) = \sum_{i=1}^{N} \underbrace{w_i \cdot \frac{(\boldsymbol{\Sigma}\boldsymbol{w})_i}{\sigma(\boldsymbol{w})}}_{\mathrm{RC}_i}, \tag{31}$$

where RC_i is the **risk contribution** of asset i and $(\boldsymbol{\Sigma}\boldsymbol{w})_i = \sum_j \Sigma_{ij} w_j$ is the **marginal risk**.

The **percentage risk contribution** is:

$$\%\mathrm{RC}_i = \frac{w_i \cdot (\boldsymbol{\Sigma}\boldsymbol{w})_i}{\boldsymbol{w}^\top \boldsymbol{\Sigma}\, \boldsymbol{w}}. \tag{32}$$

7.2 ERC Formulation (Maillard–Roncalli–Teïlétche)

The Equal Risk Contribution (ERC) portfolio [8] requires $\mathrm{RC}_i = \mathrm{RC}_j$ for all pairs (i, j):

$$\min_{\boldsymbol{w}} \sum_{i=1}^{N} \sum_{j=1}^{N} \Big(w_i \, (\boldsymbol{\Sigma} \boldsymbol{w})_i - w_j \, (\boldsymbol{\Sigma} \boldsymbol{w})_j \Big)^2, \quad \text{s.t.} \quad \boldsymbol{w}^\top \mathbf{1} = 1, \; w_i \geq 0. \tag{33}$$

7.3 Log-Barrier Formulation (Spinu)

The objective in (33) is a quartic polynomial in $\boldsymbol{w}$ and is *not convex*—gradient-based solvers may converge to local minima. This motivates the reformulation below.

Spinu [9] showed that ERC is equivalent to the strictly convex problem:

$$\boxed{\min_{\boldsymbol{x} > \mathbf{0}} \quad \frac{1}{2} \boldsymbol{x}^\top \boldsymbol{\Sigma} \, \boldsymbol{x} - \sum_{i=1}^{N} b_i \, \ln x_i,} \tag{34}$$

where $b_i = 1/N$ for ERC (or b_i equals the desired risk budget for general risk budgeting), and $\boldsymbol{w} = \boldsymbol{x} / \|\boldsymbol{x}\|_1$ after solving.

Proposition 7.1 (Strict Convexity)**.** *The Hessian of* (34) *is* $\boldsymbol{H} = \boldsymbol{\Sigma} + \mathrm{diag}(b_i / x_i^2)$*. Since* $\boldsymbol{\Sigma}$ *is PSD and* $\mathrm{diag}(b_i / x_i^2)$ *is strictly positive definite for* $\boldsymbol{x} > \mathbf{0}$*, the problem has a unique global minimum.*

7.4 Relaxed Risk Parity Fallback

When $\boldsymbol{\Sigma}$ is near-singular (common during high-correlation crypto bear markets), the standard solver may fail. The *Relaxed Risk Parity* (RRP) formulation recasts the problem as a second-order cone program (SOCP):

$$\begin{aligned} \min_{\boldsymbol{w}, \psi, \gamma} \quad & \psi - \gamma \\ \text{s.t.} \quad & \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \leq \psi^2, \\ & w_i \, (\boldsymbol{\Sigma} \boldsymbol{w})_i \geq \gamma^2 b_i \quad \forall \, i, \\ & \boldsymbol{w}^\top \mathbf{1} = 1, \; \boldsymbol{w} \geq \mathbf{0}, \; \psi, \gamma \geq 0. \end{aligned} \tag{35}$$

The objective drives ψ down (less total risk) and γ up (more equal contributions). When the strict ERC solution exists, RRP recovers it exactly.

8 Mean-CVaR Optimization

8.1 Value-at-Risk and Conditional Value-at-Risk

Definition 8.1 (Value-at-Risk)**.** *For a confidence level* $\alpha \in (0, 1)$*, the VaR of a portfolio with return* R_p *is:*

$$\mathrm{VaR}_\alpha = -\inf \Big\{ x \in \mathbb{R} : F_{R_p}(x) > 1 - \alpha \Big\}, \tag{36}$$

i.e., the α*-quantile loss. Equivalently,* $\Pr(R_p < -\mathrm{VaR}_\alpha) \leq 1 - \alpha$*. For continuous return distributions, this simplifies to* $\mathrm{VaR}_\alpha = -F_{R_p}^{-1}(1 - \alpha)$*.*

Definition 8.2 (Conditional Value-at-Risk (Expected Shortfall))**.**

$$\mathrm{CVaR}_\alpha = -\mathbb{E}\Big[R_p \mid R_p \le -\mathrm{VaR}_\alpha\Big] = \frac{1}{1-\alpha}\int_{-\infty}^{-\mathrm{VaR}_\alpha} (-x)\, f_{R_p}(x)\, dx. \tag{37}$$

CVaR is a *coherent risk measure* [11], satisfying sub-additivity ($\mathrm{CVaR}(\boldsymbol{w}_1 + \boldsymbol{w}_2) \le \mathrm{CVaR}(\boldsymbol{w}_1) + \mathrm{CVaR}(\boldsymbol{w}_2)$), whereas VaR is not.

8.2 Rockafellar–Uryasev LP Reformulation

Rockafellar and Uryasev [10] introduced the auxiliary function:

$$F_\alpha(\boldsymbol{w}, \zeta) = \zeta + \frac{1}{1-\alpha}\,\mathbb{E}\Big[\max\Big(0,\ -\boldsymbol{r}^\top \boldsymbol{w} - \zeta\Big)\Big], \tag{38}$$

where $\zeta \in \mathbb{R}$ is an auxiliary variable.

Theorem 8.1 (Rockafellar–Uryasev, 2000)**.** *1.* $\mathrm{CVaR}_\alpha(\boldsymbol{w}) = \min_{\zeta\in\mathbb{R}} F_\alpha(\boldsymbol{w}, \zeta)$.
2. The minimizer ζ^* *equals* $\mathrm{VaR}_\alpha(\boldsymbol{w})$.
3. F_α *is jointly convex in* $(\boldsymbol{w}, \zeta)$.

For S historical return scenarios $\boldsymbol{r}_1, \ldots, \boldsymbol{r}_S$ (each equally likely), introduce auxiliary variables $u_s \ge 0$:

$$\boxed{\begin{aligned} \min_{\boldsymbol{w},\zeta,\boldsymbol{u}} \quad & \zeta + \frac{1}{S(1-\alpha)}\sum_{s=1}^{S} u_s \\ \text{s.t.} \quad & u_s \ge -\boldsymbol{r}_s^\top \boldsymbol{w} - \zeta, \quad s = 1, \ldots, S, \\ & u_s \ge 0, \quad s = 1, \ldots, S, \\ & \boldsymbol{w}^\top \mathbf{1} = 1, \ \ 0 \le w_i \le w_{\max}. \end{aligned}} \tag{39}$$

This is a **linear program** with $N + 1 + S$ decision variables and $2S$ inequality constraints, solvable in polynomial time.

9 Black–Litterman Model

The Black–Litterman model [3] combines a market-equilibrium prior with investor views via Bayesian inference to produce stable posterior return estimates.

9.1 Equilibrium Prior

If the market-capitalization-weighted portfolio $\boldsymbol{w}_{\mathrm{mkt}}$ is mean-variance optimal, the implied equilibrium excess returns are:

$$\boldsymbol{\pi} = \delta\, \boldsymbol{\Sigma}\, \boldsymbol{w}_{\mathrm{mkt}}, \tag{40}$$

where $\delta = (\mu_m - r_f)/\sigma_m^2$ is the market risk-aversion coefficient. The prior distribution on expected returns is:

$$\boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\pi},\ \tau\boldsymbol{\Sigma}), \tag{41}$$

where $\tau > 0$ (typically 0.025–0.05) scales the prior uncertainty.

9.2 View Specification

An investor expresses K views through:

- **Pick matrix** $\boldsymbol{P} \in \mathbb{R}^{K\times N}$: each row encodes one view as a linear combination of asset returns.
- **View vector** $\boldsymbol{Q} \in \mathbb{R}^{K}$: the expected return for each view.
- **Uncertainty matrix** $\boldsymbol{\Omega} \in \mathbb{R}^{K\times K}$: diagonal matrix of view variances. Using the He–Litterman specification [4]:

$$\boldsymbol{\Omega} = \operatorname{diag}\Big(\tau\, \boldsymbol{P}\, \boldsymbol{\Sigma}\, \boldsymbol{P}^{\top}\Big). \tag{42}$$

  In practice, $\boldsymbol{\Omega}$ is taken as the *diagonal* of $\tau\boldsymbol{P}\boldsymbol{\Sigma}\boldsymbol{P}^{\top}$, assuming independent view errors—the standard implementation in riskfolio-lib.

View types.

- *Absolute view* on asset i: row of $\boldsymbol{P}$ has $+1$ at position i, zeros elsewhere.
- *Relative view* (asset i outperforms j by q): row has $+1$ at i, -1 at j.

9.3 Bayesian Posterior

Applying Bayes' theorem with Gaussian prior (41) and Gaussian likelihood $\boldsymbol{P}\boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{Q}, \boldsymbol{\Omega})$:

$$\boxed{\mathbb{E}[\boldsymbol{R}] = \Big[(\tau\boldsymbol{\Sigma})^{-1} + \boldsymbol{P}^{\top}\boldsymbol{\Omega}^{-1}\boldsymbol{P}\Big]^{-1}\Big[(\tau\boldsymbol{\Sigma})^{-1}\boldsymbol{\pi} + \boldsymbol{P}^{\top}\boldsymbol{\Omega}^{-1}\boldsymbol{Q}\Big].} \tag{43}$$

The posterior covariance is:

$$\boldsymbol{\Sigma}_{\text{post}} = \Big[(\tau\boldsymbol{\Sigma})^{-1} + \boldsymbol{P}^{\top}\boldsymbol{\Omega}^{-1}\boldsymbol{P}\Big]^{-1}. \tag{44}$$

An equivalent form, derived via the Woodbury matrix identity, avoids inverting the $N \times N$ matrix $\tau\boldsymbol{\Sigma}$ and instead inverts the $K \times K$ matrix $\boldsymbol{P}\tau\boldsymbol{\Sigma}\boldsymbol{P}^{\top} + \boldsymbol{\Omega}$, where $K \ll N$ (number of views $\ll$ number of assets):

$$\mathbb{E}[\boldsymbol{R}] = \boldsymbol{\pi} + \tau\boldsymbol{\Sigma}\boldsymbol{P}^{\top}\Big[\boldsymbol{P}\,\tau\boldsymbol{\Sigma}\,\boldsymbol{P}^{\top} + \boldsymbol{\Omega}\Big]^{-1}(\boldsymbol{Q} - \boldsymbol{P}\boldsymbol{\pi}). \tag{45}$$

9.4 On-Chain Signal Views

In our implementation, views are derived from DeFiLlama on-chain data:

1. **TVL momentum** $> 5\%$: generates a relative view that ETH outperforms BTC.
2. **Stablecoin inflow** $> 3\%$: generates an absolute bullish view on all assets equally.
3. **DEX volume surge** $> 30\%$: generates an absolute view favoring DeFi tokens (from the candidate set {UNI, AAVE, MKR, LINK}, filtered to those present in the current universe).

Each signal is mapped to a row of $\boldsymbol{P}$ and an element of $\boldsymbol{Q}$, with confidence encoded through $\boldsymbol{\Omega}$. When no signals are triggered, the model falls back to Maximum Sharpe Markowitz optimization.

10 Hidden Markov Model Regime Detection

10.1 Model Definition

A Gaussian Hidden Markov Model [13, 14] is defined by the triplet $\lambda = (\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{\rho})$:

Definition 10.1 (Gaussian HMM).

- ***Hidden states***: $\mathcal{S} = \{s_1, \ldots, s_K\}$ *with* $K \in \{2, 3\}$.
- ***Transition matrix***: $\boldsymbol{A} \in \mathbb{R}^{K \times K}$, *where* $A_{ij} = \Pr(q_t = s_j \mid q_{t-1} = s_i)$.
- ***Emission distributions***: $b_j(o_t) = \mathcal{N}(o_t;\, \mu_j, \sigma_j^2)$ *for each state* j.
- ***Initial distribution***: $\rho_j = \Pr(q_1 = s_j)$.

10.2 Forward Algorithm

The forward variable $\alpha_t(j)$ computes the joint probability of the observations up to time t and being in state j:

$$\alpha_1(j) = \rho_j \cdot b_j(o_1), \tag{46}$$

$$\alpha_t(j) = b_j(o_t) \sum_{i=1}^{K} \alpha_{t-1}(i)\, A_{ij}, \quad t = 2, \ldots, T. \tag{47}$$

The model likelihood is $\Pr(\boldsymbol{O} \mid \lambda) = \sum_{j=1}^{K} \alpha_T(j)$.

10.3 Backward Algorithm

The backward variable $\beta_t(i)$ computes the probability of future observations given the current state:

$$\beta_T(i) = 1, \tag{48}$$

$$\beta_t(i) = \sum_{j=1}^{K} A_{ij}\, b_j(o_{t+1})\, \beta_{t+1}(j), \quad t = T{-}1, \ldots, 1. \tag{49}$$

10.4 Baum–Welch Algorithm (EM)

The Baum–Welch algorithm [14] estimates λ from observations by iterating:

E-Step. Compute posterior state probabilities:

$$\gamma_t(j) = \frac{\alpha_t(j)\, \beta_t(j)}{\sum_{k=1}^{K} \alpha_t(k)\, \beta_t(k)}, \tag{50}$$

$$\xi_t(i, j) = \frac{\alpha_t(i)\, A_{ij}\, b_j(o_{t+1})\, \beta_{t+1}(j)}{\Pr(\boldsymbol{O} \mid \lambda)}. \tag{51}$$

M-Step. Re-estimate parameters:

$$\hat{\rho}_j = \gamma_1(j), \tag{52}$$

$$\hat{A}_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}, \tag{53}$$

$$\hat{\mu}_j = \frac{\sum_{t=1}^{T} \gamma_t(j)\, o_t}{\sum_{t=1}^{T} \gamma_t(j)}, \tag{54}$$

$$\hat{\sigma}_j^2 = \frac{\sum_{t=1}^{T} \gamma_t(j)\, (o_t - \hat{\mu}_j)^2}{\sum_{t=1}^{T} \gamma_t(j)}. \tag{55}$$

10.5 Viterbi Algorithm

The Viterbi algorithm [15] finds the most likely hidden state sequence via dynamic programming:

$$\delta_1(j) = \rho_j \cdot b_j(o_1), \tag{56}$$

$$\delta_t(j) = \max_{i \in \{1,\ldots,K\}} \Big[\delta_{t-1}(i) \cdot A_{ij}\Big] \cdot b_j(o_t), \tag{57}$$

$$\psi_t(j) = \arg\max_{i \in \{1,\ldots,K\}} \Big[\delta_{t-1}(i) \cdot A_{ij}\Big]. \tag{58}$$

The optimal final state is $q_T^* = \arg\max_j \delta_T(j)$, and the full sequence is recovered by backtracking: $q_t^* = \psi_{t+1}(q_{t+1}^*)$.

Remark 10.1 (Log-Space Implementation)**.** *In practice, all computations use log-probabilities to prevent floating-point underflow:* $\ln \delta_t(j) = \max_i[\ln \delta_{t-1}(i) + \ln A_{ij}] + \ln b_j(o_t)$.

10.6 State Identification

State labels from Baum–Welch are arbitrary. Regimes are identified by emission means after fitting:

$$\text{Bull} = \arg\max_{j \in \{1,\ldots,K\}} \mu_j, \qquad \text{Bear} = \arg\min_{j \in \{1,\ldots,K\}} \mu_j. \tag{59}$$

For $K = 3$, the remaining state is labeled "Sideways."

11 Regime-Aware Allocation

Following Ang and Bekaert [16], the regime-aware strategy selects an optimization strategy conditional on the current market regime:

$$\boldsymbol{w}_t = \begin{cases} \boldsymbol{w}_{\text{MaxSharpe}} & \text{if } q_T^* = \text{Bull}, \\ \boldsymbol{w}_{\text{MinVar}} & \text{if } q_T^* = \text{Bear}, \\ \boldsymbol{w}_{\text{ERC}} & \text{if } q_T^* = \text{Sideways}. \end{cases} \tag{60}$$

This is a *hard switching* rule based on the Viterbi state at time T. An alternative *soft-blending* approach, not implemented in the current system but proposed for future work, uses the posterior state probabilities from the forward algorithm:

$$\boldsymbol{w}_t^{\text{soft}} = \sum_{j=1}^{K} \gamma_T(j)\, \boldsymbol{w}^{(j)}, \tag{61}$$

which would produce smoother transitions between strategies at the cost of higher turnover near regime boundaries.

12 Risk and Performance Metrics

12.1 Core Performance Ratios

Definition 12.1 (Sharpe Ratio [2]).

$$S = \frac{\mu_{ann} - r_f}{\sigma_{ann}}. \tag{62}$$

Definition 12.2 (Sortino Ratio [19]).

$$So = \frac{\mu_{ann} - r_f}{\sigma_{down} \cdot \sqrt{365}}, \qquad \sigma_{down} = \sqrt{\frac{1}{T-1} \sum_{t:\, r_t < 0} r_t^2}. \tag{63}$$

Definition 12.3 (Calmar Ratio [20]).

$$C = \frac{\mu_{ann}}{|MDD|}. \tag{64}$$

Definition 12.4 (Omega Ratio [21]). *For threshold $\tau = 0$:*

$$\Omega = \frac{\sum_{t:\, r_t > 0} r_t}{|\sum_{t:\, r_t < 0} r_t|} = \frac{\mathbb{E}[\max(R, 0)]}{\mathbb{E}[\max(-R, 0)]}. \tag{65}$$

12.2 Drawdown Analysis

Definition 12.5 (Maximum Drawdown). *Let V_t be the portfolio equity curve. The drawdown at time t is:*

$$DD(t) = \frac{V_t - \max_{s \le t} V_s}{\max_{s \le t} V_s} = \frac{V_t}{HWM(t)} - 1, \tag{66}$$

where $HWM(t) = \max_{s \le t} V_s$ is the running high-water mark. The maximum drawdown is:

$$MDD = \min_{t \in [1,T]} DD(t). \tag{67}$$

12.3 Tail Risk Measures

For empirical (historical simulation) estimation with T return observations:

$$\mathrm{VaR}_\alpha = -\hat{F}^{-1}(1-\alpha) = -\mathrm{Percentile}\Big(\{r_t\},\ (1-\alpha) \times 100\Big), \tag{68}$$

$$\mathrm{CVaR}_\alpha = -\frac{1}{|\mathcal{T}_\alpha|} \sum_{t \in \mathcal{T}_\alpha} r_t, \qquad \mathcal{T}_\alpha = \{t : r_t \le -\mathrm{VaR}_\alpha\}. \tag{69}$$

12.4 Higher Moments

$$\text{Skewness} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t - \bar{r}}{\hat{\sigma}}\right)^3, \tag{70}$$

$$\text{Excess Kurtosis} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t - \bar{r}}{\hat{\sigma}}\right)^4 - 3. \tag{71}$$

Crypto returns typically exhibit negative skewness (crash risk) and large positive excess kurtosis (5–15, vs. ≈ 0.5 for equities).

13 Walk-Forward Backtesting

13.1 Methodology

The walk-forward backtester evaluates strategies without look-ahead bias using a rolling training window.

Definition 13.1 (No Look-Ahead Guarantee)**.** *At each rebalance date τ_k, the weight vector is computed using only past data:*

$$\boldsymbol{w}_{\tau_k} = f\Big(\{r_t\}_{t=\tau_k-L}^{\tau_k-1}\Big), \tag{72}$$

where L is the lookback window length (default: 365 days) and f is the chosen optimization strategy.

13.2 Rebalancing Mechanics

Let $\{\tau_1, \tau_2, \ldots, \tau_K\}$ be the set of rebalance dates (weekly, monthly, or quarterly). Between rebalances, weights drift with market returns:

$$w_{i,t}^{\text{drift}} = \frac{w_{i,t-1}\,(1 + r_{i,t-1})}{\sum_{j=1}^{N} w_{j,t-1}\,(1 + r_{j,t-1})}. \tag{73}$$

In the implementation, $r_{i,t}$ are log returns; the expression $(1+r_{i,t})$ approximates $e^{r_{i,t}}$. For daily returns with $|r| < 0.05$, the approximation error is $\mathcal{O}(r^2) < 0.13\%$.

13.3 Daily Portfolio Return

$$R_{p,t} = \sum_{i=1}^{N} w_{i,t}\, r_{i,t} = \boldsymbol{w}_t^{\top}\boldsymbol{r}_t. \tag{74}$$

13.4 Transaction Cost Model

At each rebalance τ_k, the one-way turnover and cost are:

$$\text{Turnover}_k = \sum_{i=1}^{N}\Big|w_{i,\tau_k}^{\text{new}} - w_{i,\tau_k}^{\text{drift}}\Big|, \tag{75}$$

$$\text{TC}_k = \text{Turnover}_k \times \frac{c}{10{,}000}, \tag{76}$$

where c is the cost in basis points (default: 10 bps). The portfolio value is reduced before applying new weights:

$$V_{\tau_k}^{\text{post}} = V_{\tau_k} \times (1 - \text{TC}_k). \tag{77}$$

13.5 Equity Curve Construction

Starting from initial capital V_0:

$$V_t = V_{t-1} \times (1 + R_{p,t}), \quad t = 1, 2, \ldots, T. \tag{78}$$

14 Summary of Strategies

Table 1 summarizes the seven allocation strategies and their key mathematical properties.

Table 1: Comparison of portfolio optimization strategies.

Strategy	**Requires Σ^{-1}**	**Requires μ**	**Convex**	**Eq. ref**
Equal Weight	No	No	N/A	(18)
Markowitz MVO	Yes	Yes	QP	(7)
GARCH-Enhanced GMV	Yes	No	QP	(14), (28)
HRP	No	No	N/A	Alg. 1
Risk Parity (ERC)	No	No	Yes	(34)
Mean-CVaR	No	Yes	LP	(39)
Black–Litterman	Yes	Yes	QP	(43)
Regime-Aware	Var	Var	Var	(60)

N/A indicates the strategy is not formulated as an optimization problem: Equal Weight is a closed-form rule, and HRP is a greedy recursive algorithm.

15 Conclusion

This paper has presented the complete mathematical framework underlying our multi-strategy cryptocurrency portfolio optimization system. The diversity of approaches—from classical mean-variance theory to machine-learning-based regime detection—provides complementary perspectives on the risk–return trade-off in crypto markets. Each strategy addresses specific limitations of the others: HRP mitigates MVO's sensitivity to estimation error, CVaR captures tail risk that variance ignores, Black–Litterman incorporates on-chain fundamental signals, and regime-aware allocation adapts dynamically to changing market conditions. The walk-forward backtesting framework enables rigorous, bias-free comparison of all strategies under realistic transaction cost assumptions.

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